



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

TECHNICAL MATHEMATICS P1

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Technical Mathematics Paper 1 practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept correct alternative methods; give follow-through marks (CA) where an earlier error is used correctly.

QUESTION 1

- 1.1 $\sqrt{9} = 3$: rational ✓; π : irrational ✓; $-\frac{3}{4}$: rational ✓; $\sqrt{-4}$: non-real ✓ (4)
- 1.2 $45 = 32 + 8 + 4 + 1$ ✓ $= 101101_2$ ✓ (2)
- 1.3 $16 + 8 + 0 + 2 + 0$ ✓ $= 26$ ✓ (2)
- 1.4 $1011_2 + 110_2 = 10001_2$ ✓✓ $(11 + 6 = 17)$ (2)
- 1.5 3,142 ✓ (1)
- 1.6 $4,5$ ✓ $\times 10^{-5}$ ✓ (2)
- 1.7 $\sqrt{49} = 7$ and $\sqrt{64} = 8$, so between 7 ✓ and 8 ✓ (2)

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QUESTION 2

- 2.1.1 $8a^9b^6$ ✓✓ (2)
- 2.1.2 $= \frac{x^3}{x^2}$ ✓✓ $= x$ ✓ (3)
- 2.1.3 $4x^4$ ✓✓ (2)
- 2.1.4 $2\sqrt{(4 \times 3)} + \sqrt{(9 \times 3)} = 4\sqrt{3}$ ✓ $+ 3\sqrt{3}$ ✓ $= 7\sqrt{3}$ ✓ (3)
- 2.2.1 $(x - 3)(x - 4)$ ✓✓ (2)
- 2.2.2 $(2x - 5)(2x + 5)$ ✓✓ (2)
- 2.2.3 $3x(x + 2)$ ✓✓ (2)
- 2.3 $= \frac{(x - 2)(x + 2)}{x + 2}$ ✓✓ $= x - 2$ ✓ (3)
- 2.4 $2x^2 - 3x + 8x - 12$ ✓✓ $= 2x^2 + 5x - 12$ ✓ (3)
- 2.5 $6x - 3 - 2x - 8$ ✓✓ $= 4x - 11$ ✓ (3)

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QUESTION 3

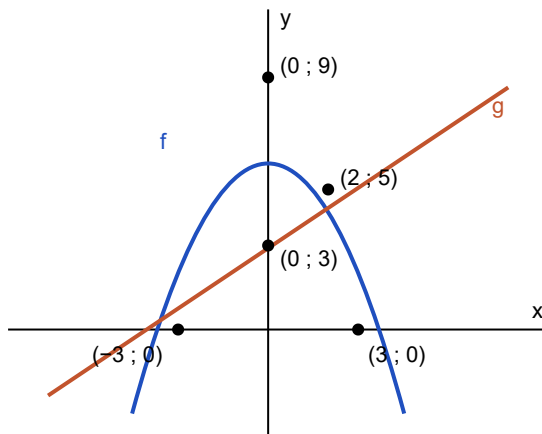
- 3.1.1 $3x = 12$ ✓ $\rightarrow x = 4$ ✓ (2)
- 3.1.2 $(x - 3)(x + 2) = 0$ ✓ $\rightarrow x = 3$ ✓ or $x = -2$ ✓ (3)
- 3.1.3 $2^x = 2^5$ ✓ $\rightarrow x = 5$ ✓ (2)

- 3.2 $-2x > 4 \checkmark \rightarrow x < -2 \checkmark$ (sign reverses when dividing by a negative);
number line with an open circle at -2 and arrow to the left $\checkmark$ (3)
- 3.3 $3x + (2x + 1) = 11 \checkmark \rightarrow 5x = 10 \checkmark \rightarrow x = 2 \checkmark$; $y = 2(2) + 1 = 5 \checkmark$ (4)
- 3.4 $R = \frac{V}{I} \checkmark = \frac{12}{0,5} \checkmark = 24 \Omega \checkmark$ (3)
- 3.5 $30 = 2(9 + b) \checkmark \rightarrow 15 = 9 + b \checkmark \rightarrow b = 6 \text{ cm} \checkmark$ (3)
- 3.6 Let width = x : $x(x + 3) = 40 \checkmark \rightarrow x^2 + 3x - 40 = 0 \checkmark \rightarrow (x + 8)(x - 5) = 0$
 $\checkmark$
 $x = 5$ (width cannot be negative) $\checkmark$; width 5 cm and length 8 cm $\checkmark$ (5)

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QUESTION 4

- 4.1 $(0 ; 9) \checkmark$ (1)
- 4.2 $-x^2 + 9 = 0 \rightarrow x = \pm 3 \checkmark$; $(-3 ; 0)$ and $(3 ; 0) \checkmark$ (2)
- 4.3 $x + 3 = 0 \rightarrow (-3 ; 0) \checkmark$ (1)
- 4.4 $x = 0 \checkmark$; gradient = 1 $\checkmark$ (2)
- 4.5 (5)



Marks: shape of f (opens downwards) $\checkmark$; turning point $(0 ; 9) \checkmark$; x-intercepts of $f \checkmark$; straight line g with positive gradient $\checkmark$; intercepts of g : $(0 ; 3)$ and $(-3 ; 0) \checkmark$

- 4.6 $-x^2 + 9 = x + 3 \rightarrow x^2 + x - 6 = 0 \checkmark \rightarrow (x + 3)(x - 2) = 0 \checkmark$
 $(-3 ; 0) \checkmark$ and $(2 ; 5) \checkmark$ (4)
- 4.7 $y \leq 9 \checkmark$ (1)
- 4.8 $-3 \leq x \leq 2 \checkmark \checkmark$ (2)
- 4.9 $x = 0 \checkmark$ and $y = 0 \checkmark$; quadrants 1 and 3 $\checkmark$ (3)
- 4.10 $(0 ; 1) \checkmark$; $y = 0 \checkmark$ (2)
- 4.11 $y = x - 2 \checkmark \checkmark$ (2)

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QUESTION 5

5.1 (a) $A = P(1 + i)^n = 15\,000(1,09)^4 \checkmark\checkmark = R21\,173,72 \checkmark\checkmark$ (7)

(b) $A = P(1 + in) = 15\,000(1 + 0,09 \times 4) \checkmark\checkmark = R20\,400 \checkmark$

5.2 $21\,173,72 - 20\,400 = R773,72 \checkmark$ (1)

5.3 $A = 40\,000(1 - 0,1 \times 3) \checkmark = R28\,000 \checkmark$ (2)

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TOTAL: 100

END OF MARKING GUIDELINE